MINSUK JUN

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RESEARCH INTERESTS

Continuous-variable photonic quantum computing, generation of GKP states through cat state breeding, and low-loss χ(2) photonic integrated circuits on thin-film lithium tantalate (LTOI).

EDUCATION

From 03/27 — **Integrated M.S./Ph.D. in Electrical and Computer Engineering** — Seoul, KR
Seoul National University (admitted), Q-SPIN Lab, Advisor: Prof. Donguk Nam

2020 - 2027 — **B.S. in Electrical and Electronic Engineering** — Suwon, KR
Sungkyunkwan University (SKKU), expected graduation Feb 2027
GPA 4.20 / 4.5
- Coursework: Quantum Mechanics, Electromagnetics, Optics, Linear Algebra

08/25 - 12/25 — **Exchange Student, Nanyang Technological University (NTU)** — Singapore
GPA 5.0 / 5.0
- Top of class in Semiconductor Devices and Processing
- Coursework in semiconductor physics, optoelectronics and device fabrication

RESEARCH EXPERIENCE

06/26 - Present — **Research Intern, Q-SPIN Lab** — Seoul, KR
Seoul National University, Advisor: Prof. Donguk Nam
Continuing in the lab as an integrated M.S./Ph.D. student from March 2027
- Work on the theory side of the lab's LTOI chip (cat state generation, cat breeding, GKP), setting the loss and squeezing each component on the chip must reach.
- Design beam splitters and pump filters for the tape-out, comparing MMI, directional and adiabatic couplers and 775/1550 nm pump rejection with eigenmode (FDE) simulation in Ansys Lumerical MODE.
- Simulate photonic quantum circuits, including photon-pair sources and MZI networks in Perceval and loss-aware Gaussian-state models of GKP generation.
- Presented graph-designed quantum photonics (Bao et al., Nature Photonics 2023) at the lab group meeting.

02/26 - 06/26 — **Undergraduate Research Assistant** — Suwon, KR
Prof. Juwon Lee's Lab (Semiconductor Device & Materials Lab), Sungkyunkwan University
- Studied 2D transition metal dichalcogenides (TMDs) for optoelectronics and presented a review of MoS2-based memory devices.
- Took part in weekly group seminars on device fabrication and characterization (Raman spectroscopy, photoluminescence).

03/25 - 07/25 — **Undergraduate Research Assistant (URP)** — Suwon, KR
Prof. Tae-wook Kang's Lab, Department of Semiconductor and Display Engineering, SKKU
- Selected for the Undergraduate Research Program and worked in a four-person team on an integrated circuit for real-time noise cancellation in voice signals.
- Designed the FFT core of an 8-bit STFT module in Verilog, choosing the fixed-point quantization from Python statistics of the signal values.

LEADERSHIP AND SERVICE

12/24 - 07/25 — **Vice President, SIOR (SKKU Institute of Robot)** — Suwon, KR
- Ran technical seminars for club members and mentored students in robotics fundamentals.
- Organized an Arduino outreach program at Yeongbok Girls' High School, teaching assembly and programming.

2022 - 2024 — **Military Service, Republic of Korea Air Force**
- Honorably discharged as Sergeant.

SKILLS

Simulation: Ansys Lumerical MODE (FDE), Perceval, The Walrus, PSpice
Programming: Python (NumPy, SciPy), Verilog, C/C++, MATLAB, Arduino
Languages: Korean (native), English, Japanese (JLPT N2)